Redistributing Human Work in Robotic-Assisted Photogrammetry and Focus Stacking for Small Museum Object Digitisation

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Abstract

Object digitisation workflows such as photogrammetry and focus stacking require many overlapping images and careful coordination across imaging processes. Robotic arms can provide repeatable multi-view positioning, while focus stacking extends depth of field for macro imaging, but integrating these systems does not necessarily reduce human effort. This exploratory study examines four robotic-assisted image-acquisition sessions involving three small museum artefacts using a collaborative robotic arm, motorised turntable, macro rail, and camera system. Rather than treating the setup as a fully automated pipeline or controlled benchmark, we analyse how work was redistributed across the workflow. Robotic positioning reduced repeated manual camera placement, but human judgement remained necessary for calibration, focal adjustment, object reorientation, capture verification, masking, reconstruction review, and quality assessment. We therefore characterise robotic-assisted digitisation as a partially automated socio-technical workflow and derive design considerations for making calibration, handoff, monitoring, data organisation, and output assessment more legible and recoverable.

CCS Concepts

• **Computing methodologies**; • **Computer systems organisation**; • **Architectures**;

Keywords

Robotic-assisted photogrammetry, Focus stacking, Object digitisation, Human–machine interaction

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1 INTRODUCTION

High-resolution three-dimensional digitisation of physical artefacts is increasingly used across cultural heritage documentation [19], design prototyping [10], and museum collections [16]. Photogrammetry has become a widely adopted method for producing detailed 3D models from overlapping two-dimensional (2D) images because it can be implemented with comparatively accessible imaging equipment. However, successful photogrammetry depends on more than image acquisition alone. Practitioners must coordinate artefact preparation, lighting, camera positioning, image overlap, scale, masking, reconstruction settings, and post-processing. These demands become more pronounced when digitising small objects, where shallow depth-of-field (DOF), occluded surfaces, fine surface detail, and object handling introduce additional workflow complexity [15, 17].

Robotic systems offer one way to support repeatable image capture without relying entirely on fixed camera rigs, multi-camera arrays, or object-specific custom setups [3, 21, 22]. A robotic arm can provide precise and repeatable camera positioning, enable controlled multi-view capture, and reduce inconsistencies introduced by manual photography. Focus stacking can also support macro-scale digitisation by combining multiple focal slices into images with extended depth of field [8]. Combining robotic positioning, turntable rotation, focus stacking, and photogrammetric reconstruction may therefore improve capture control, but it also introduces a more complex workflow spanning several devices and software environments.

However, greater capture control does not necessarily produce a simpler or fully automated workflow. The operators must coordinate several semi-automated components and determine what

the system has done, what occurs next, and when intervention is needed [4, 13]. Drawing on work regarding automation, situated action [14, 24], articulation work [1, 20] [2], intelligibility, and accountability, this paper examines robotic-assisted digitisation as a socio-technical workflow organised around calibration, handoff, monitoring, data management, and output assessment.
This paper asks:

RQ1: How does robotic assistance redistribute human work across small-object digitisation workflows?

RQ2: Where do interaction challenges emerge in calibration, handoff, monitoring, data management, and output assessment?

RQ3: What design considerations can inform future robotic-assisted digitisation systems that make these workflows more legible, recoverable, and usable in practice?

Using a collaborative robotic arm and macro-imaging setup, we examined two situated workflow configurations: standard robotic-assisted photogrammetric capture and robotic-assisted focus-stacked capture. The workflows were applied to different artefacts and were not designed as directly controlled experimental conditions. We therefore compare how human and machine work was configured within each case, rather than attributing observed differences solely to the imaging method.

This paper contributes (i) an exploratory account of human and machine work across four robotic-assisted small-object digitisation sessions; (ii) characterises how partial automation redistributed human work toward anticipation, coordination, monitoring, and recovery, including articulation work around calibration, cross-device handoffs, data organisation, and output assessment; and (iii) develops design considerations for making these partially automated workflows more legible, traceable, and recoverable.

2 METHODOLOGY

2.1 Research Approach

This study adopts an exploratory research approach [7] to investigate where human intervention remained necessary in robotic-assisted photogrammetry and focus-stacked capture for small-object digitisation. The study uses the implementation and operation of a robotic-assisted imaging workflow as a situated case through which interaction issues can be identified, rather than treating system development itself as the principal research contribution. Instead of benchmarking technical performance or evaluating a finished system, we examined the workflow as a socio-technical process involving calibration, monitoring, system handoff, data organisation, post-processing, and output assessment.

2.2 System Setup

As shown in Figure 1, the experimental setup consisted of a Universal Robots UR10 collaborative robotic arm, a Canon 1300D camera mounted on a Cognisys StackShot macro rail [23], and a Cognisys actuated turntable [6]. The robotic arm was controlled through Grasshopper [9] and provided gross camera positioning, while the macro rail produced fine linear displacement along the optical axis for focus-stacking. The system was configured for artefacts approximately 2-5 cm in size.

Three artefacts were digitised across four image-acquisition sessions: the Red Dragon, Pot, and Sarcophagus. Standard robotic-assisted photogrammetry was used for the Pot and Sarcophagus, while focus-stacked capture was used for the Red Dragon. Because each workflow was applied to different artefacts and used workflow-specific acquisition parameters, the sessions were treated as situated cases rather than matched experimental conditions. The analysis consequently examines differences in workflow configuration and human involvement without attributing them exclusively to the use or absence of focus stacking.

Because the camera, rail, and mounting hardware formed a relatively large and offset end effector, robotic path planning had to account for clearance and collision risks involving the robot, artefact, turntable, and surrounding surfaces. These spatial constraints were recorded as part of the workflow analysis because they contributed to calibration, monitoring, and operator intervention.

2.3 Imaging Workflows

Both (i) standard photogrammetry capture and (ii) focus-stacked capture followed the same broad sequence: setup and calibration, image acquisition, multi-orientation capture, post-processing, and output evaluation, as depicted in Figure 2. This shared structure provided a common analytical frame for describing how human and machine work was distributed in each workflow. The comparison is descriptive rather than experimental because the workflows involved different artefacts, rotational increments, image volumes, and processing requirements.

2.3.1 Setup and Calibration. The turntable centre and rotational axis were registered relative to the robot base frame using pose data accessed through the real-time data exchange (RTDE) protocol. This supported spatial registration of the turntable centre relative to the robotic base frame (see Figure 3). This reference supported repeatable camera positioning around the artefact. The focal target was defined from the approximate artefact centre, and the tool centre point of the imaging end effector was calibrated before capture.

2.3.2 Image Acquisition Workflows. In standard photogrammetry, one image was captured at each rotational position using 5° turntable increments. In focus-stacked capture, multiple focal slices were acquired at each position and combined into extended-depth-of-field images before reconstruction [8]. At each robotic position, multiple images were captured at different focal distances. The number of focal slices was also adjusted according to the camera elevation because the apparent depth of the artefact relative to the sensor decreased. Focus-stacked capture used 10° turntable increments to offset the additional acquisition time.

Both workflows used camera elevations of approximately 0°, 20°, 40°, and 60°. To improve geometric coverage, image acquisition was generally undertaken across at least two object orientations. During the initial capture sequence, the object was imaged in its upright position. It was then manually repositioned to expose surfaces occluded during the first pass, particularly the underside and support-contact regions. A subsequent capture sequence was performed using the same robotic imaging procedure.

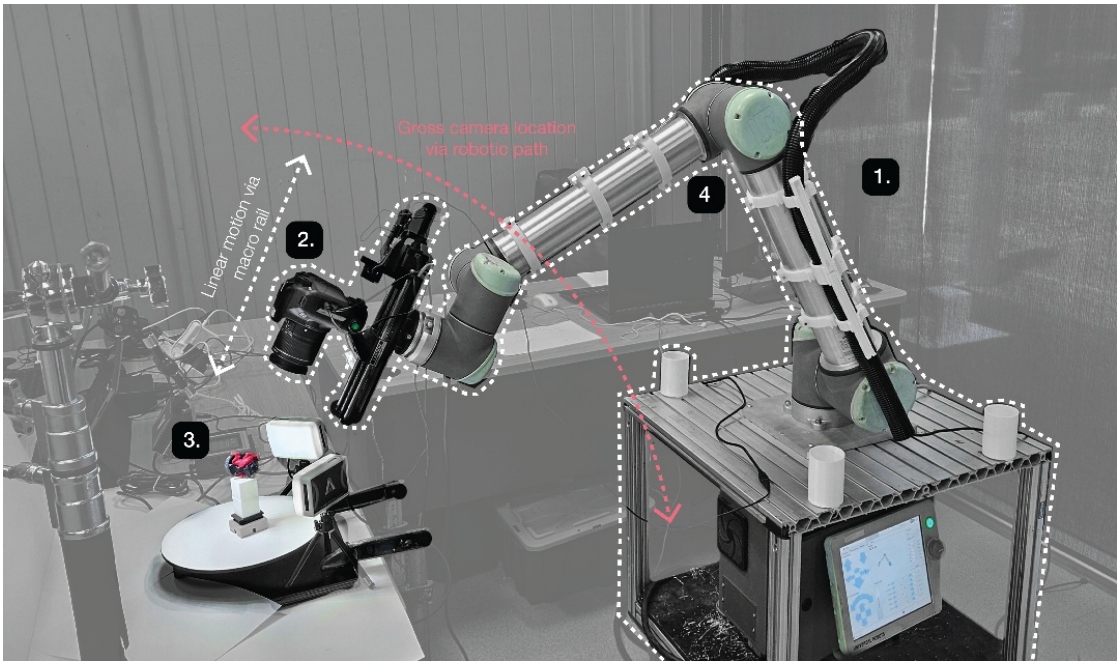


Figure 1: Experimental robotic-assisted digitisation setup, showing the lighting arrangement, backdrop, (1) collaborative robotic arm with (2) camera end-effector, artefact, (3) motorised rotational table, and Windows PC (4) used for controlled image acquisition.

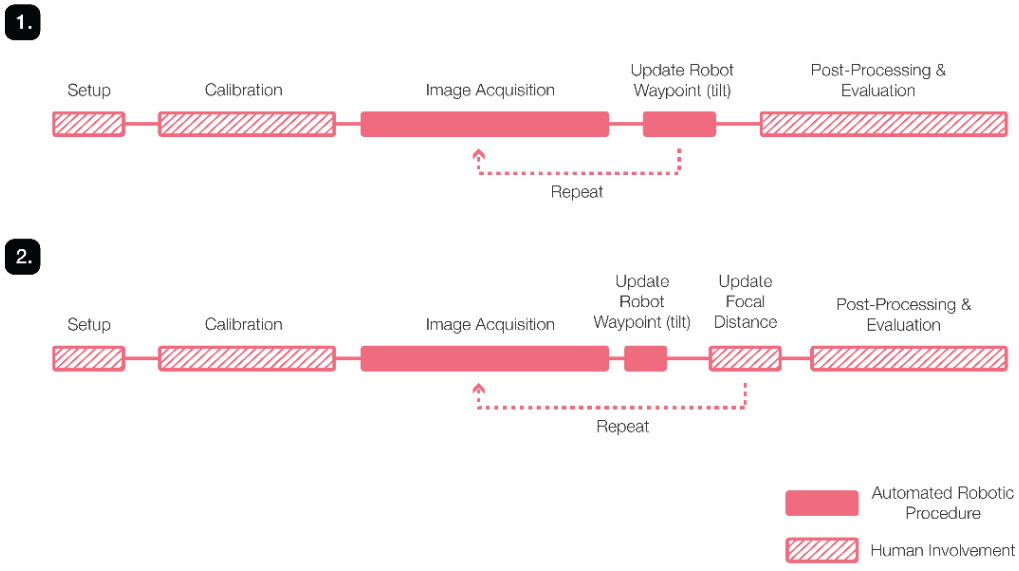


Figure 2: Overall robotic-assisted digitisation workflow. The process moves from physical setup and calibration to image acquisition, multi-orientation capture, post-production, and output evaluation. The workflow shows how both (1) standard robotic-assisted photogrammetry and (2) focus-stacked capture depend on coordinated preparation, robotic positioning, image processing, and expert review of final output quality.

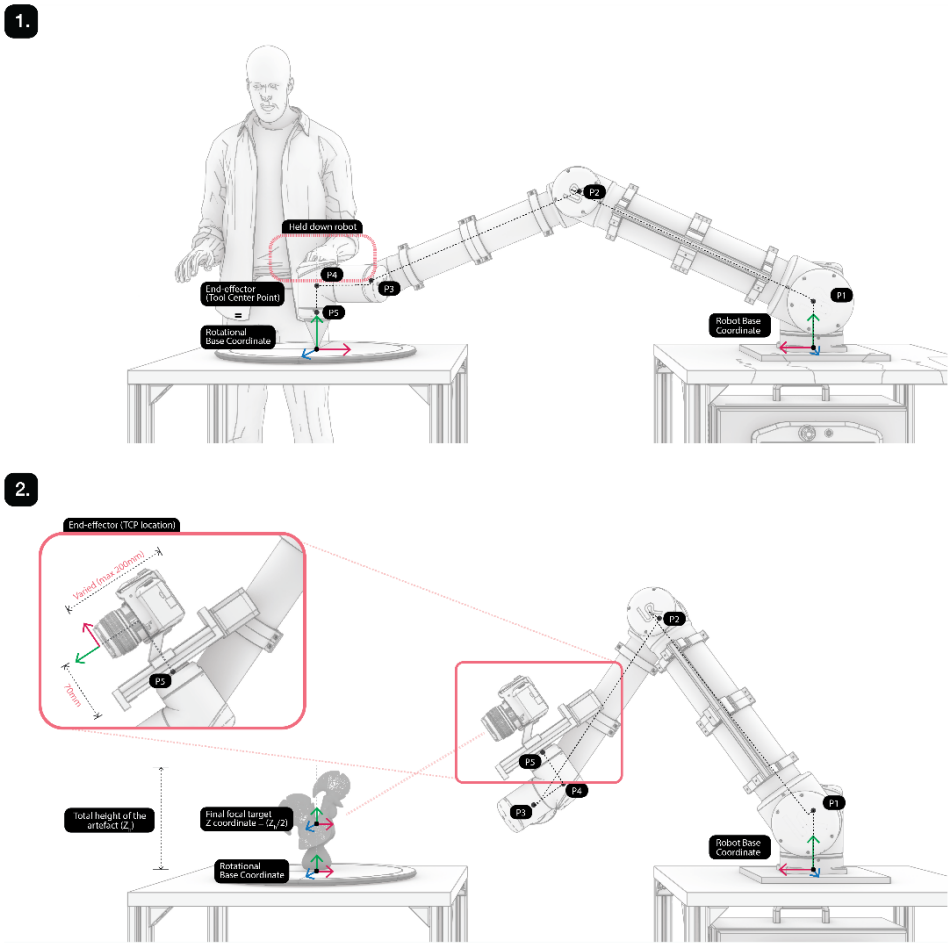


Figure 3: Coordinate relationship between the robotic arm, rotational table, and camera end-effector. The robot base coordinate frame is calibrated in relation to the centre and rotational axis of the turntable, while the camera lens position is defined relative to the robot tool flange. This calibration establishes the spatial relationship required to program repeatable camera viewpoints around the artefact.

2.3.3 Post-production Workflows and Output Evaluation. Post-production included image review, focus-stacked generation where applicable, masking, photogrammetric reconstruction in RealityScan [18], mesh merging, and output review. Separate orientation datasets were masked and merged to improve coverage and reduce duplicated or occluded geometry.

The outputs from both workflows were evaluated through expert assessment and supporting geometric comparison. Researcher 2, who had expertise in image acquisition and digitisation workflows, reviewed the resulting image sets and reconstructed models to assess artefact coverage, visibility of surface detail, reconstruction of occluded geometry, and suitability for the intended documentation purpose. Where CT-derived reference meshes were available, the photogrammetric reconstructions were compared against them using a procedure previously applied in related geometric-comparison studies [12, 13]. The meshes were first manually pre-aligned, then refined using iterative closest point registration, before mesh-to-mesh surface distances were calculated [25]. This comparison was used as secondary supporting evidence for interpreting geometric deviation, missing coverage, and reconstruction incompleteness.

2.4 Data Collection

Data collection focused on both imaging outputs and workflow characteristics. For each image-acquisition session, we recorded image count, capture duration, dataset size, processing time, calibration effort, and post-processing effort. Image-acquisition duration was estimated from the elapsed time between the EXIF timestamps of the first and final images, including necessary setup adjustments during acquisition.

Human interventions were documented through field notes, process photographs, and observational records produced by the first and second authors, who were also the primary operators of the robotic-assisted digitisation workflows examined in this study. The

first author primarily configured and operated the robotic system, while the second author contributed expertise in image acquisition and digitisation and participated in capture, workflow monitoring, and output assessment. These records captured moments where manual adjustment, troubleshooting, or interpretive judgement was required, including calibration refinement, object repositioning, focal adjustment, capture supervision, masking, dataset alignment, reconstruction review, and output quality assessment. Intervention counts were interpreted in context rather than as directly equivalent metrics, because focus-stacked capture required routine focal adjustment that varied with camera elevation, artefact geometry, and focal-depth range.

2.5 Data Analysis Procedure and Researchers' Position

The analysis used a hybrid deductive–inductive approach informed by Braun and Clarke's [5] thematic analysis procedure. Initial deductive attention was directed toward forms of work commonly associated with partially automated systems, including calibration, coordination, monitoring, intervention, and recovery. Within these broad areas, the researchers examined the observational material inductively to identify recurring patterns specific to the robotic-assisted digitisation workflows.

The qualitative dataset comprised contemporaneous field notes, process photographs, and observational records produced during the four image-acquisition sessions. These materials documented manual adjustments, troubleshooting activities, coordination between researchers and system components, and instances where interpretive judgement was required. The analysis also drew on EXIF-derived acquisition timings, image and dataset characteristics, reconstructed outputs, and expert assessment of output adequacy.

The first and second authors initially reviewed the field notes, process photographs, and observational records separately. Relevant incidents were coded according to the type of work undertaken, the stage of the workflow in which it occurred, the reason intervention was required, and the consequence for subsequent capture or reconstruction. The researchers then compared their interpretations and grouped related codes into candidate themes. Through discussion and iterative review against the source material, these themes were refined into five recurring interaction issues: calibration, system handoff, supervisory monitoring, data management, and output assessment.

These themes were therefore not treated as mutually exclusive categories or as quantitative measures of effort. Rather, they were used to characterise where and how human work remained necessary across the partially automated workflow. Intervention timings were also interpreted in relation to the circumstances in which they occurred, rather than treated as directly equivalent measures across the two workflows.

Findings were corroborated across multiple forms of evidence. Field notes and observational records were compared against process photographs, acquisition timings, reconstructed outputs, and expert review. Where CT-derived reference meshes were available, mesh-to-mesh comparison provided additional evidence regarding any geometric variation and incomplete coverage. These comparisons were interpreted as secondary supporting evidence rather than as standalone measures of absolute reconstruction accuracy.

The research team was directly involved in configuring, operating, documenting, and assessing the system. Their position provided detailed access to situated decisions and breakdowns, but also meant that the analysis reflected practitioner-researcher interpretations rather than independent user accounts. To address this limitation, observations were compared across researchers and evidence sources, and differences in interpretation were discussed within the research team. The analysis was also informed by complementary expertise in photography, digital preservation, robotics, and human–robot interaction.

3 RESULTS

This section reports findings from the two robotic-assisted image-acquisition workflows, focusing on capture time, data volume, calibration, system handoff, operator intervention, and quality judgement. Across four acquisition sessions, robotic assistance supported programmable and repeatable camera positioning, consistent with prior robotic scanning research [11], but did not remove the need for human involvement. Instead, human effort shifted toward setup, monitoring, coordination, data management, post-processing, and assessment of output suitability.

3.1 Partial Automation Produced Different Configurations of Human Work

The observed workflow cases involved different capture durations, data volumes, and patterns of human intervention. These differences describe the particular configurations examined in this study and should not be interpreted as effects attributable solely to standard photogrammetry or focus stacking. The standard photogrammetry workflow required approximately 1 hour and 30 minutes of image-acquisition time and used 5° rotational increments. This workflow was comparatively faster during image acquisition and produced a smaller dataset. However, the shorter capture duration did not mean that the overall workflow was simpler. The process still required careful attention to image overlap, object visibility, turntable rotation, camera angle, and reconstruction adequacy.

Focus-stacked capture required approximately 5 hours and 30 minutes because multiple focal slices were acquired at each viewpoint and processed before reconstruction. It also required additional configuration of the macro rail and imaging assembly. To partially offset this burden, focus-stacked capture used 10° rotational increments, producing 36 final stacked images per elevation, compared with the denser 5° sampling used for standard photogrammetry.

As summarised in Table 1, the observed cases involved different configurations of work. In the focus-stacked case, work was concentrated around calibration, focal sequencing, and data organisation. In the standard photogrammetry cases, more visible effort concerned coverage judgement, operator coordination, and reconstruction review. These differences are analytically useful for identifying forms of human involvement, but they cannot be separated from differences in artefact geometry, acquisition parameters, and session-specific decisions.

Table 1: Descriptive profile of the observed robotic-assisted digitisation workflow cases, including capture duration, data volume, processing requirements, intervention points, and output characteristics. Because the workflows were applied to different artefacts and acquisition configurations, the table does not represent a controlled performance comparison.

Workflow dimension	Focus-Stacked Capture	Standard Photogrammetry
Approximate capture time	Around five hours and thirty minutes	Around one hour and thirty minutes
Approximate image count	Higher; multiple slices per viewpoint; 1,980 original and 288 processed photographs	Lower; one image per viewpoint; 720 and 616 photographs respectively
Approximate dataset size	Larger due to source slices and stacked output	Smaller image set
Post-processing demand (include mesh stitching)	Focus-stack generation, reconstruction, mesh repair, mesh merging and external software transfer	Image review, masking, alignment, reconstruction review, and mesh merging
Primary output	Extended-depth-of-field images used for digital reconstruction	Image set used directly for reconstruction
Main capture requirement	Stable framing across focal movement	Sufficient angular coverage and image overlap
Main intervention points	Calibration, focus range adjustment, view verification, file organisation	Coverage judgement, operator coordination, reconstruction review
Main workflow risk	Cropped or inconsistent focal stack	Incomplete coverage or reconstruction failure.
Expert assessment	Supported extended depth of field and visibility of surface detail under macro-imaging conditions, but remained dependent on focal-stack quality and reconstruction settings.	Potentially constrained by shallow depth of field under macro-imaging conditions, particularly for geometrically deep or irregular surfaces.

3.2 Calibration became Anticipatory Human Work

We found that initial calibration was critical in both workflows. The robotic arm, camera, turntable, and artefact had to be spatially aligned so that the artefact remained visible throughout the programmed sequence. This required registering the robot coordinate system relative to the turntable centre and positioning the artefact within a reliable camera–object relationship.

In standard photogrammetry, calibration primarily affected coverage. Poor object placement or camera alignment could produce gaps, insufficient overlap, or views unsuitable for reconstruction. In focus-stacked capture, calibration was more sensitive because the camera translated along the optical axis during each focal sequence. An inaccurate setup could cause the object to become cropped, shift between slices, or approach the camera assembly too closely. The operator therefore had to anticipate the entire range of rail movement rather than verify only the initial frame.

Furthermore, focal-stack parameters were adjusted across robotic elevation angles. As camera elevation increased, the apparent depth of the artefact relative to the camera sensor decreased, allowing fewer focal slices to be used. Determining an appropriate focal range therefore required the operator to interpret the relationship between camera orientation, artefact geometry, and the depth range requiring focus coverage. This reduced capture time at higher elevations but introduced a situated judgement task within the programmed sequence.

Once established, robotic positioning reduced repeated manual camera placement and object-specific rigging. However, the reliability of the sequence depended on the initial calibration. Successful calibration allowed capture to proceed with limited interruption, whereas setup errors were propagated across multiple viewpoints and images.

3.3 Handoff Gaps Exposed Fragmented Automation

The two workflows differed in how transitions between capture stages were coordinated. In the focus-stacked workflow, the macro rail and turntable were synchronised through the StackShot controller. At the end of each focal sequence, the rail activated a microswitch that triggered the robotic arm to move to the next programmed elevation. This enabled one mechanical action to initiate the next without direct human coordination. However, the handoff was only partially automated. After the robot moved, the operator verified that the artefact remained within the camera frame and adjusted the focal range where necessary, as shown in Figure 4.

In the standard photogrammetry workflow, the transition between turntable capture and robotic repositioning was not machine-to-machine. Researcher 2 received the completion command from the StackShot controller and informed Researcher 1, who then moved the robot to the next elevation. The handoff therefore depended on human-to-human communication to bridge components that were not directly integrated. These observations show that automation was unevenly distributed across the workflow and strongly shaped by whether hardware components could communicate directly.

3.4 Data Management and Reconstruction Review Remained Human Judgement Tasks

As highlighted earlier, the focus-stacked workflow generated a substantially larger dataset because each final reconstruction image was derived from multiple focal slices. This increased the volume of source data that had to be named, stored, transferred, checked, processed, and organised before photogrammetric reconstruction could begin. The operator also had to maintain the relationships

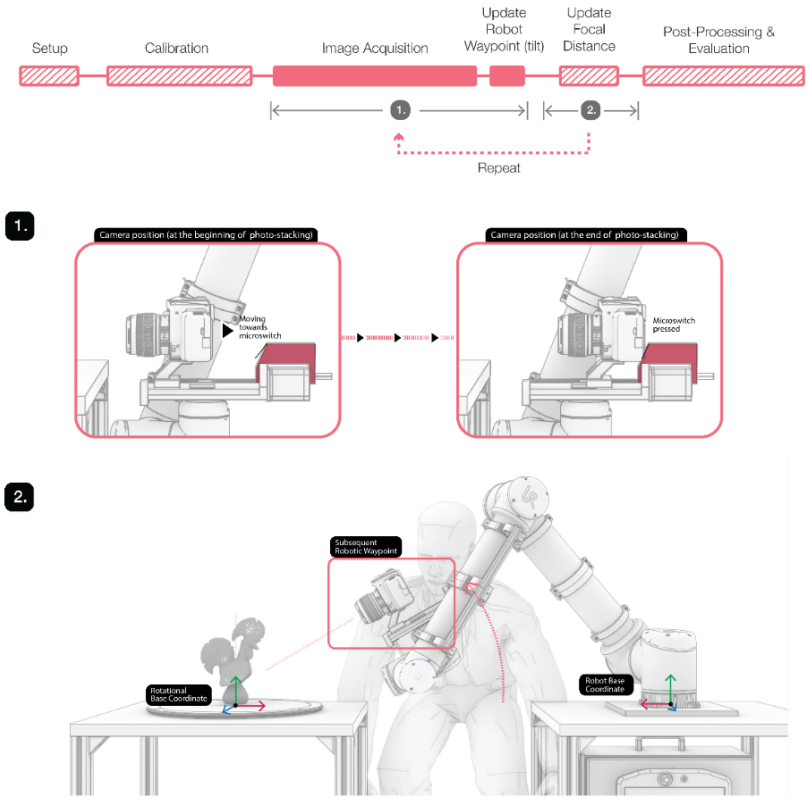


Figure 4: Human intervention in the robotic-assisted focus-stacking workflow. Operator involvement was required during initial calibration to align the robot, camera, artefact, rail, and turntable. Although the signal handoff between the rail and robotic arm was supported by a mechanical microswitch triggered when the camera rail reached the end of its travel, human involvement remained necessary to adjust focal depth for each tilt angle and verify that the artefact remained within the camera frame before continuing the next capture sequence.

between individual focal slices, processed image stacks, elevation angles, rotational positions, and the final reconstruction inputs. Data management therefore became an integral component of the digitisation workflow rather than a secondary administrative task.

The standard photogrammetry workflow generated a smaller dataset during capture, but it still required post-processing decisions. The operator had to decide whether the image set was complete enough for reconstruction and whether the resulting model required further capture, masking, alignment, or correction. In this sense, the data burden was different rather than absent.

Throughout our study, we identified that post-processing became particularly demanding when artefacts were captured in multiple orientations. Upright capture did not fully document underside and support-contact regions, so the objects were manually repositioned and captured again. The resulting datasets then required masking, alignment, and mesh merging to integrate newly visible surfaces.

These findings show that acquisition duration alone does not represent the total workload or complexity of the digitisation process. A shorter capture sequence may still produce uncertainty during reconstruction, while a longer sequence may generate substantial processing and organisational work.

3.5 Output Quality Depended on Task-Specific Judgement

Output assessment showed that reconstruction quality could not be interpreted as a single technical outcome. Instead, the usefulness of each reconstructed model depended on the (i) intended documentation task, (ii) the degree of surface coverage required, (iii) the reconstruction settings used during processing, and (iv) the amount of clean-up work that could reasonably be absorbed into the workflow.

In the standard photogrammetry workflow, the initial post-processing strategy was deliberately low intervention. All captured images were imported into RealityScan [18] without masking to test automatic alignment. This produced a near-usable reconstruction but did not fully resolve the artefact across orientations. The result indicated that additional images alone did not guarantee a

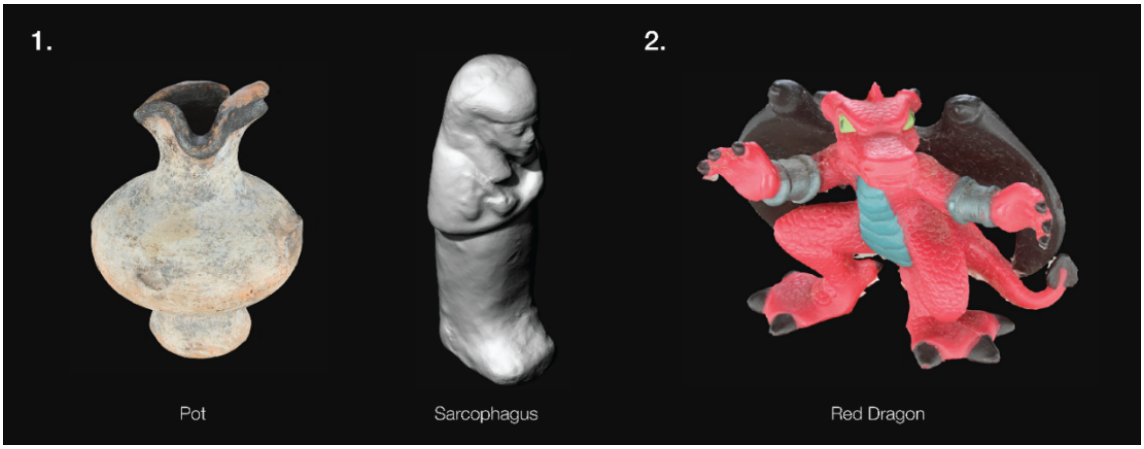


Figure 5: Representative reconstruction outputs from RealityScan. The standard photogrammetry outputs (1) illustrate usable models for viewing-oriented tasks, while the focus-stacked output (2) illustrates greater surface-detail visibility but also increased processing and review requirements.

more complete model; adequacy depended on how orientations, masks, and reconstruction settings were managed.

The upright dataset produced a visually coherent partial model (refer to Figure 5 [Left]). Expert assessment suggested that such an output could be sufficient for viewing-oriented uses, including public engagement or research contexts where underside geometry was not critical. It would be less suitable for complete archival documentation, measurement, or analysis of occluded regions.

Capture also required object-specific adaptation. An additional elevation angle was acquired for the Pot to improve confidence in geometric coverage, while the Sarcophagus underwent an additional lighting sequence to improve image suitability. These changes were not predetermined in the robotic path but resulted from expert assessment of whether the standard procedure would provide adequate visual information.

We also conducted a mesh-to-mesh comparison for the standard photogrammetry reconstructions. Following uniform rescaling and ICP registration, the Sarcophagus showed a mean signed deviation of approximately -0.012 mm (SD = 0.051 mm), while the Pot showed approximately −0.001 mm (SD = 0.204 mm) (refer to Figure 6). Because global scale was removed and positive and negative deviations may cancel in the mean, these values indicate relative surface agreement rather than absolute dimensional accuracy. Broad surface geometry was retained, although local variation and incomplete regions remained.

The focus-stacked workflow introduced additional processing and review demands because focal slices had to be processed before reconstruction. We also identified that the reconstruction settings influenced final surface quality. For example, processing the Red Dragon using the software's normal processing mode produced a smoother result, although underside geometry remained incomplete. As no CT-derived reference mesh was available, the focus-stacked output was assessed primarily through expert review. Focus stacking improved the visibility of fine surface detail, but this remained a qualitative benefit rather than a validated accuracy result. The increased detail also made defects such as holes, smoothing artefacts, and poorly reconstructed surfaces more consequential, requiring judgement about whether further repair was justified. In one case, a repair operation created a hole that could not be resolved within RealityScan, requiring export to another application before texturing could continue, as shown in Figure 7. Final output quality therefore depended on expert assessment and on the broader post-processing software ecology, rather than on image acquisition alone.

Output adequacy was therefore task-contingent, and human judgement remained necessary to determine whether a model was fit for purpose, which defects were consequential, and whether further capture or processing was justified across both workflows.

4 DISCUSSION

4.1 Automation shifted work toward anticipation, supervision, and recovery

The observed cases show that robotic positioning changed the character of human work rather than simply reducing it. Repetitive camera positioning was transferred to the robot and turntable, while human work shifted toward configuring, monitoring, and recovering the partially automated workflow.

This redistribution is significant because the remaining work was often temporally displaced from the automated action itself. Calibration decisions made before capture affected hundreds of subsequent images, while errors could remain invisible until image review or reconstruction. Human involvement was therefore concentrated at points of uncertainty: before capture, at transitions between components, and during evaluation of incomplete or defective outputs. These activities can be understood as articulation work [20], where human coordination is needed to connect otherwise fragmented tasks and technologies across the workflow. In our cases, this was particularly evident in coordinating system handoffs, maintaining relationships between datasets and reconstruction inputs, and determining when outputs were adequate to continue.

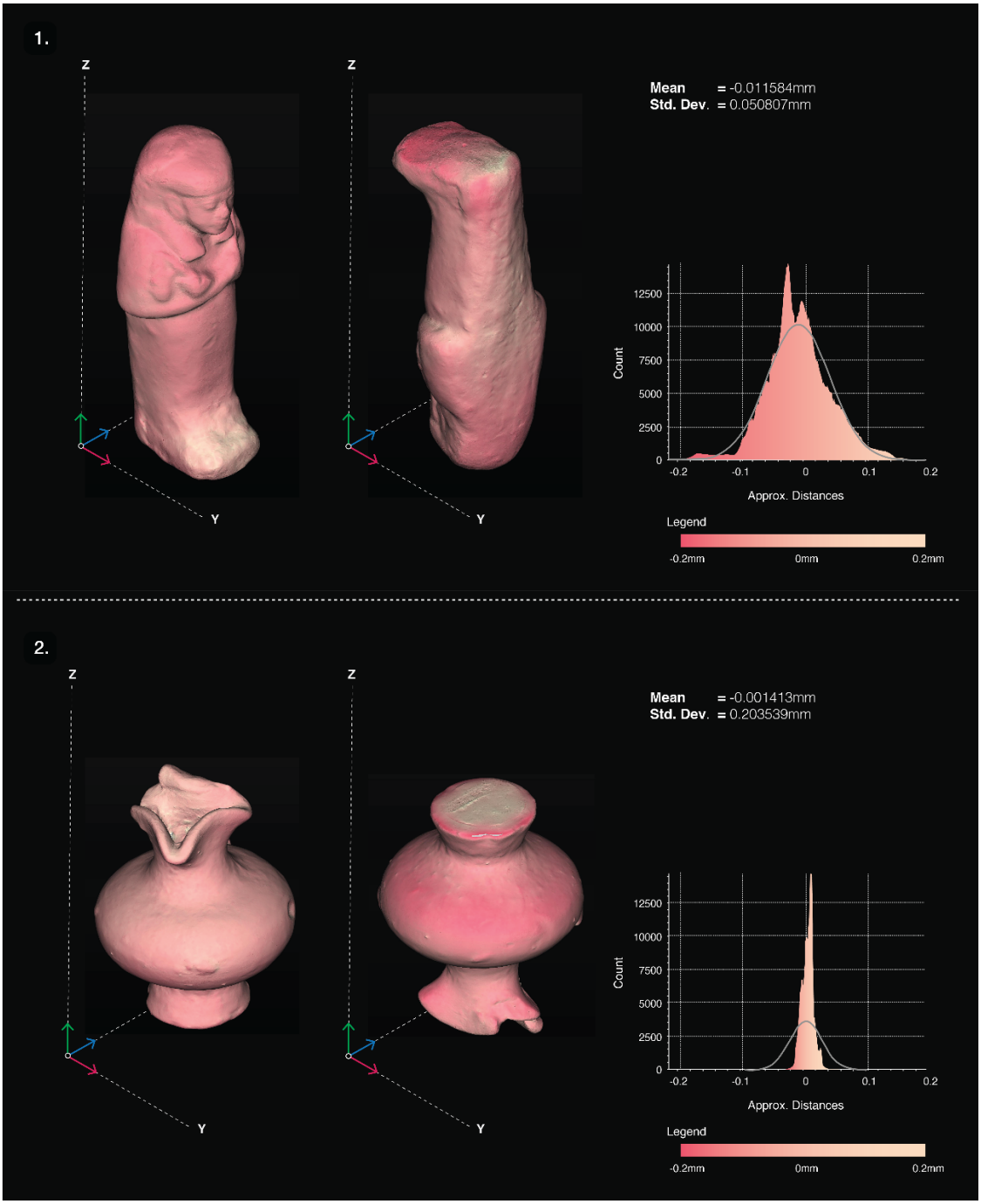


Figure 6: Supporting mesh-to-mesh comparison between the standard photogrammetry reconstructions and CT-derived reference meshes for the Sarcophagus and Pot. Deviation maps and distance distributions were generated from 100,000 sampled points following manual pre-alignment, uniform rescaling, and iterative closest point registration. The results represent relative surface agreement after alignment and scaling, not absolute dimensional accuracy.

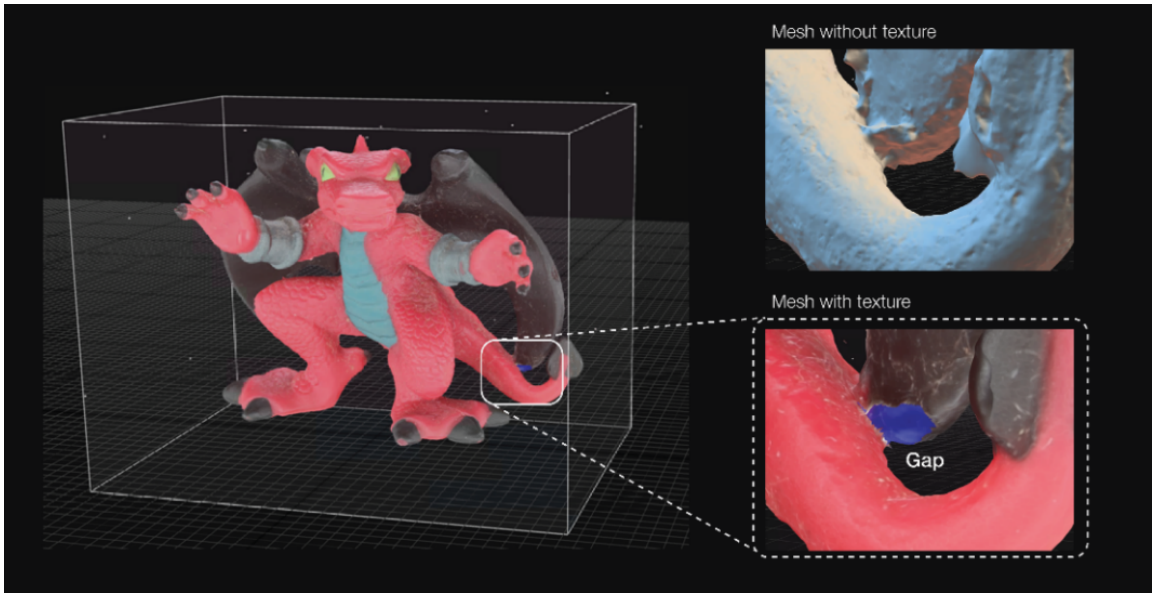


Figure 7: Mesh generated in RealityScan, showing visible reconstruction gaps in the underside geometry.

For HCI, this indicates that the value of robotic assistance cannot be assessed only through capture speed, positional repeatability, or image count. Evaluation should also consider the attentional and interpretive demands placed on operators, the cost of discovering errors late, and the support available for intervention and recovery.

4.2 Calibration and handoff are interaction design problems

Calibration functioned as a form of anticipatory interaction rather than a one-time technical adjustment. Operators had to reason about future robot poses, rail movement, focal depth, framing, and collision clearance before the consequences of those decisions became visible. Because a setup error could propagate across a long capture sequence, calibration also determined the recoverability and cost of later failures.

This reframes calibration as an HCI problem. If calibration remains opaque, users may only discover errors after a long capture sequence has failed or after reconstruction produces incomplete results. Future systems should therefore make calibration more visible, testable, and recoverable. Preview tools, field-of-view simulation, focal-range visualisation, coverage feedback, and warnings about frame loss or spatial constraints could help users assess whether the setup is viable before committing to capture.

The findings also show that usability depends on handoffs between components. Robotic-assisted digitisation often combines robotic controllers, turntables, macro rails, camera triggers, stacking software, and reconstruction software. Where these components are not directly integrated, operators must bridge transitions through timing, communication, and manual coordination.

For future systems, handoff design should be treated as a core interaction design concern. Interfaces should communicate when one stage has finished, what will occur next, whether the system is ready, and whether human confirmation is required. This is particularly important for non-specialist users who may be able to operate individual tools but struggle to diagnose failures across the broader workflow.

4.3 Workflow value depends on task goals and quality judgement

The descriptive comparison of the observed cases should not be interpreted as a controlled evaluation or as establishing a universal preference for one workflow. Standard photogrammetry and focus-stacked capture were applied to different artefacts and used different rotational increments, image volumes, and processing procedures. The findings therefore demonstrate the forms of human work that arose within each configuration, rather than isolating the causal effect of the imaging method. Each offered different benefits and operational costs.

Output quality was not meaningful independently of the intended documentation task. The same reconstruction could be adequate for visual presentation but inadequate for measurement, archival documentation, or analysis of occluded geometry. Therefore, workflow completion could not be defined solely by successful reconstruction or visual coherence.

This also makes quality assessment an interaction-design problem because the operators need support not only to produce a model, but to determine whether further capture, masking, alignment, or repair is warranted for the intended use. Future research could focus on developing interfaces that could make this judgement more explicit by linking capture coverage and reconstruction defects to task-specific quality criteria.

4.4 Design implications for legible and recoverable robotic digitisation

The particular intervention points observed in this study were shaped by our hardware and software configuration and should not be interpreted as universal requirements of robotic digitisation. We instead use these situated observations to identify broader interaction-design concerns that may recur in partially automated digitisation workflows where multiple devices, software environments, and human judgements must be coordinated. The findings suggest five design considerations for future robotic-assisted digitisation systems. First, calibration should be visible and recoverable, allowing users to verify the alignment of the robot, camera, turntable, rail, and object before committing to a capture sequence.

Second, handoffs should communicate system state, task completion, next actions, and points where human confirmation is required, particularly when devices are not controlled through a unified interface. Third, supervisory interfaces should allow operators to monitor framing, focus, coverage, capture progress, and image quality from the workstation.

Fourth, data management should be treated as part of the interaction design of the workflow. Data management should track the relationships among the generated source images, focal stacks, object-orientation datasets, masks, intermediate reconstructions, and final outputs, as the users need clear ways to track how these materials relate to one another, particularly when workflows are repeated, shared, audited, or transferred across multidisciplinary teams.

Fifth, systems should support task-specific quality assessment by helping users determine whether an output satisfies its intended documentation purpose, which defects are consequential, and whether further capture or processing is justified.

Taken together, these considerations shift the design objective from maximising automation toward making calibration, handoff, monitoring, data management, and quality assessment legible and actionable within partially automated workflows.

5 LIMITATIONS AND FUTURE WORK

This study should be understood as an exploratory workflow investigation based on a specific robotic-assisted digitisation setup. The findings were shaped by the hardware configuration, object scale, software tools, and expertise of the research team. We therefore do not claim that robotic-assisted capture is generally more efficient, accurate, or usable than conventional digitisation methods, nor that the comparison between standard photogrammetry and focus-stacked capture establishes a general performance ranking. Instead, the study identifies workflow constraints and interaction-design considerations that emerged from the implementations. Accordingly, these design considerations should be treated as propositions arising from the cases examined rather than as generalisable requirements for robotic-assisted digitisation.

Secondly, the setup was configured for small objects, limiting generalisability across artefact scales, geometries, and surface finishes. Different documentation contexts may require alternative camera distances, lenses, lighting conditions, rail travel, and robotic paths. Future work should test the workflow across a broader range of artefacts to determine how calibration, path planning, monitoring, and quality assessment should adapt to different digitisation tasks.

The study also did not include a formal user evaluation with museum practitioners, designers, or other non-specialist users, nor did it begin with a systematic formative investigation of practitioner work practices, requirements, or existing digitisation routines. Therefore, the workflow issues identified here reflect an exploratory researcher-operated implementation rather than requirements derived from a broader practitioner population. However, the second author contributed subject-matter expertise in image acquisition and digitisation workflows, providing an initial practitioner-informed perspective during system operation and output assessment. This expertise supported the exploratory investigation but should not be treated as a substitute for broader formative engagement with practitioners. Future work should therefore involve museum and digitisation practitioners in formative studies of existing work practices.

Finally, output evaluation was limited to expert assessment and supporting geometric comparison rather than a systematic benchmark across a large object set. Future work should combine workflow analysis with structured measures of reconstruction completeness, geometric deviation, surface-detail visibility, processing time, and task-specific adequacy. More integrated communication between the robotic arm, turntable, macro rail, camera trigger, focus-stacking software, and photogrammetry software should also be investigated to reduce coordination overhead and make system state more visible.

6 CONCLUSION

This exploratory study examined how robotic assistance redistributed human work across photogrammetric and focus-stacked digitisation. Although robotic positioning reduced repeated camera placement, reliable capture still depended on anticipatory calibration, cross-device coordination, supervisory verification, data organisation, and task-specific quality judgement. Calibration errors could propagate through lengthy sequences, fragmented handoffs obscured system state, and reconstruction adequacy depended on documentation purpose. Future systems should therefore prioritise calibration methodology, explicit handoffs, incremental verification, provenance support, and task-specific quality guidance.

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SUPPLEMENTARY DOCUMENTS

Supplementary materials are available in an anonymised GitHub repository https://github.com/LoyWeiWin/Pub_OZCHI_SupplementaryDocument_RedistributingHumanWorkinRAP. These include session-level spreadsheets reporting image-acquisition and

calibration durations, spreadsheets containing the point-cloud deviation data used in the mesh-to-mesh analyses of the Sarcophagus and Pot, and the Grasshopper definition and associated Rhino file for the robotic imaging setup.

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